Vegetatio **101**: 3–13, 1992.

Secondary plant succession: how is it modified by insect herbivory?

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Accepted 21.10.1991

Keywords: Insect herbivory, Root herbivore, Succession, Plant species richness

Abstract

The effects of foliar- and root-feeding insects on the dynamics of an early successional plant community, representing the first four years of colonisation, were examined. Subterranean insect herbivores were found to increase in density with increasing successional age of the plant community. In early succession, chewing insects mainly Coleoptera (Scarabaeidae) and Diptera (Tipulidae) were dominant. This was in direct contrast to the foliar-feeding insects, which were dominated by sap-feeders (mainly Auchenorrhynchan Hemiptera).

Reduction of both foliar- and root-feeding insects with appropriate insecticides had different, but dramatic, consequences for the plant community. Reducing foliar herbivory resulted in large increases in perennial grass growth, with plant species richness being reduced as the grasses outcompeted the forbs. Reducing subterranean herbivory prolonged the persistence of annual forbs, greatly increased perennial forb colonisation and, as a consequence, plant species richness. Foliar-feeding insects thus act to delay succession by slowing grass colonisation. In contrast, root-feeding insects accelerate succession by reducing forb persistence and colonisation. The structure of early successional plant communities is therefore modified by the two modes of herbivory.

Introduction

In recent years, there has been an increasing awareness of the importance of interactions between different trophic levels in the structuring of communities (e.g. Lawton & Strong 1981). In addition to theoretical and descriptive field studies, manipulative field experiments have considerable potential for a better understanding of the mechanisms which underlie the organisation of communities (Harper 1977). It is therefore encouraging that they are becoming a more prominent feature of ecological methodology (e.g. Louda 1982; Kinsman & Platt 1984; Brown *et al.* 1988). Thus, in addition to plant-soil and plant-plant interactions, those between plants and vertebrate or invertebrate herbivores and plants and pathogens are being explored to varying degrees (Gibson *et al.* 1987; Brown & Gange 1989a; Paul *et al.* 1989; Prins & Nell 1990). This paper focuses on interactions between herbivorous insects and plants, and aims to demonstrate that insect herbivory may be an important determinant in the structuring of successional plant communities.

The dynamics of these interactions are likely to change in different habitats. Whereas, in mature habitats, the association between various types of organism is generally stable, in ephemeral or rud-

This paper was presented at the Vth International Congress of Ecology (INTECOL), Japan, 1990, entitled 'Successional Communities of Plants and Insects'.

eral habitats the nature of the interactions may change rapidly. Such a range in habitat stability is well displayed along a secondary successional gradient, where a ruderal community quickly changes into an early successional community which then changes less rapidly to more stable mid and late successional communities. It is hardly surprising, though it has seldom been demonstrated, that the insect fauna closely tracks these successional changes in the plant community (Southwood *et al.* 1979). Generally, the number of insect species and the abundance of individuals increases with successional age. Of more significance to the functioning of communities is the relative contribution of the various feeding categories or guilds (Root 1973). Hendrix *et al.* (1988) have considered the insect guild structure of early successional communities in the New and Old World, and have compared the contribution of the herbivores (both chewing and sap-feeding species), parasitoids, predators, scavengers and detritivores in terms of numerical abundance. The herbivores are the most common guild, with the sap-feeders being numerically dominant, though contributing rather less to insect biomass. The contribution of the herbivores declines with successional age, while the predators and parasitoids become increasingly important (see Brown & Southwood 1987). In the ruderal community, the herbivores are mainly forb (non-graminaceous species) feeders, while in the early and mid successional communities grass feeders are dominant.

While the above-ground insect fauna associated with early successional plant communities is reasonably well documented (e.g. Waloff 1980; Brown & Hyman 1986; Brown & Southwood 1987), much less is known about the nature and composition of the insect fauna below ground. Indeed, community ecologists have generally ignored the subterranean fauna (but see Brown & Gange 1990). This lack of investigation most likely stems from difficulties with the sampling and taxonomy of soil-dwelling insects and the lack of appropriate methodology to examine ecological interactions below ground.

However, when one examines the insect species found associated with successional plant communities above ground, (e.g. Hendrix *et al.* 1988), it is immediately apparent that many species have immature stages living in the soil. Indeed, many beetle species, such as Curculionoidea (weevils) and Scarabaeidae (chafers) and the Diptera (Tipulidae), have root-feeding larvae which spend a large proportion of their life cycle below ground. Thus, no study of plant-insect interactions is complete, or even really meaningful, without the consideration of the below-ground herbivores.

The relatively large herbivore loads associated with the foliage and probably roots of successional plant communities poses the question – what effect(s), if any, does the insect community have on these rapidly changing plant assemblages? The aim of this paper is to describe briefly the nature of the below-ground insect fauna in early succession and to compare the effects these insects have on plant communities with those of foliar-feeding insects. Finally, some of the mechanisms underlying the effects are explored.

Materials and methods

An experimental site, within a large area of arable land at Silwood Park, Berkshire, U.K., was treated with a weedkiller in autumn to destroy perennial weeds and shallow ploughed in winter. The land was harrowed and hand-raked in March and the vegetation left to develop naturally. There were four treatments: (i) application of soil insecticide to reduce levels of below-ground herbivory, (ii) application of foliar insecticide to reduce levels of insects associated with foliar and reproductive structures, (iii) application of both compounds to afford a complete reduction in herbivory and (iv) control (natural levels of root- and foliar-feeding insect herbivores). Each treatment was replicated five times in a randomised block design. A replicate was a 3 m × 3 m plot, separated from any other by a 2 m barrier zone. The soil insecticide was Dursban 5G (containing 5% w/w chlorpyrifos (Dow Elanco)) applied at the rate of 18 g per plot at six-weekly intervals.

The foliar insecticide was Dimethoate-40 (Portman Agrochemicals Ltd) applied at the recommended agricultural rate (Martin & Worthing 1976) of 0.336 kg a.i. ha^{-1} (equivalent to 750 ml per plot) at three-weekly intervals from May to October. Control plots received equal volumes of water. The site was fenced to exclude rabbits and pellets of 'Mifaslug' (Farmers Crop Chemicals Ltd) were applied to reduce mollusc herbivory.

The developing vegetation was sampled at fortnightly intervals from April to October in the first season (1986), (representing a ruderal plant community), and at monthly intervals in three subsequent seasons (1987–89), (representing an early successional community). A 38 cm linear steel frame, containing ten 3 mm diameter point quadrat pins, was randomly placed five times in each plot. The number of touches of all living plant material was recorded in 2 cm (below 10 cm) or 5 cm (10 cm and above) height intervals. Data were condensed to provide information on the number of plant species (species richness), the total touches of all vegetation (cover abundance) and the total touches of plant species belonging to different life-history groupings (e.g. annual and perennial forbs and grasses (Gibson *et al.* 1987)). Data were analysed according to dates with a split-plot Analysis of Variance (ANOVA), with the two pesticide treatments as main effects.

The below-ground insect fauna was sampled in the ruderal site during the first and fourth years after establishment, and, for comparative purposes, in older sites of 5, 10 and 20 years old, representing early, early-mid and mid successional respectively. Samples were taken at six-weekly intervals in each season. Three 15 cm × 10 cm diameter soil cores were taken from each site, hand sorted to remove larger invertebrates and then placed in Tullgren funnels for two weeks (see Edwards & Fletcher 1971).

Results

Below-ground insects

Figure 1a shows the density of soil-dwelling herbivorous insects in February of the first season. The numbers of insects per m^2 increased with the successional age of the site. However, sampling confirmed the findings of other workers, namely that these insects are very heterogeneously distributed; a feature which resulted in relatively large standard errors. The sample from the experimental site four years after establishment (= early successional) demonstrates the composition of the herbivores (Fig. 1b). Chewing insects, most notably larvae of Coleoptera (Elateridae and Scarabaeidae) and Diptera (Tipulidae), were dominant while aphids were the most common below-ground sap-feeding insects.

Effects of insect herbivores on plant community characteristics

The application of foliar insecticide caused a significant increase in the cover abundance of the vegetation during each of the first four years of succession (Fig. 2; Table 1). The effect was even evident in the fourth year, despite an extremely dry summer which caused vegetative growth to be greatly reduced (Fig. 2d). Application of soil insecticide had a similar effect to that of the foliar treatment, with a significant increase in cover in each year (Fig. 2; Table 1). In the first year, however, the effect of soil insecticide was not as pronounced as that of the foliar compound (Fig. 2a). The compounds, when applied together, appeared to have a solely additive effect, with no significant interaction terms in the ANOVA (Table 1).

During the first year of succession, differences in the total vegetation cover were largely a result of the growth of annual forbs (Fig. 3a), with both compounds increasing the abundance of this life-history grouping. In subsequent years, the contribution of this group to the plant community declined rapidly in control and treatment plots (Fig. 3). As a result, there was no effect of application of the foliar insecticide on this group after the first year. However, application of soil insecticide significantly increased the cover abundance of the annual forbs in all subsequent seasons (Table 1), and both treatments which contained this compound showed a marked increase in annual forb growth (Fig. 3).

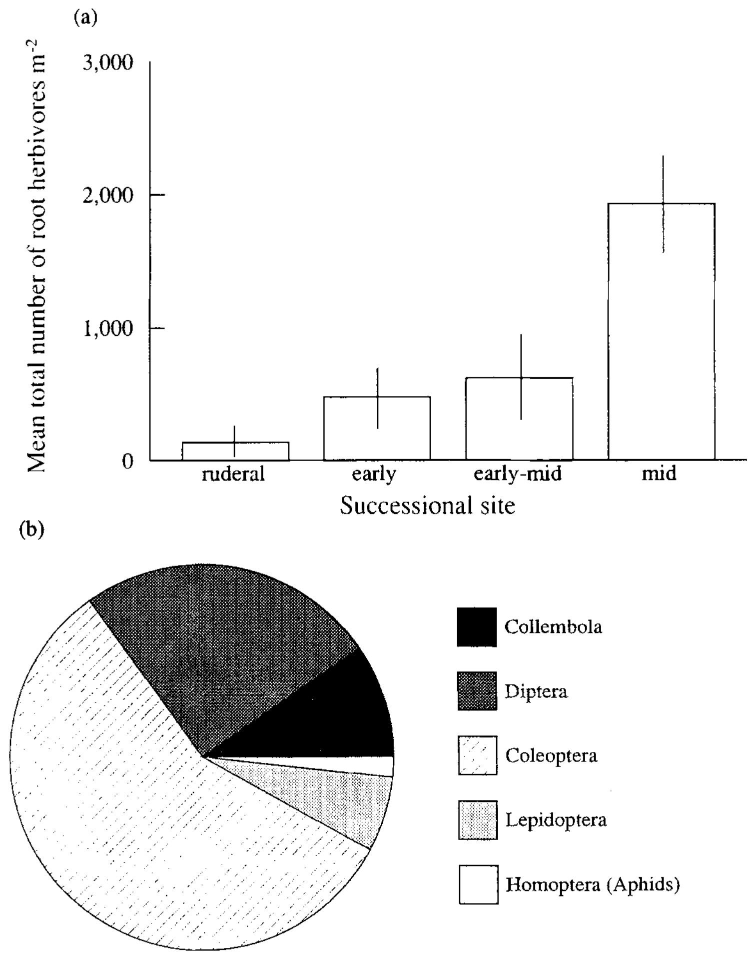

Fig. 1. (a) Changes in the density of subterranean insect herbivores with successional age of a plant community (see text). Vertical lines represent one standard error; (b) The composition of the below-ground herbivore community four years after plant establishment.

Table 1. Summary of ANOVA F values comparing plant community characteristics resulting from a reduction in above- (F) or below-ground (S) insect herbivory. Degrees of freedom for all F values = 1,16. Significance: * p < 0.05, ** p < 0.01, *** p < 0.001.

Community characteristics	Year 1			Year 2			Year 3			Year 4		
	F	S	F * S	F	S	F * S	F	S	F * S	F	S	F * S
Total cover abundance (c.a.)	57.6***	11.4**	0.7	35.3***	35.8***	1.3	96.9***	78.6***	0.6	8.9**	15.2**	0.2
Annual forbs c.a.	53.2***	10.7**	0.8	1.6	6.1*	1.2	1.1	26.6***	0.7	0.5	6.9*	1.3
Perennial forbs c.a.	0.5	1.5	0.1	0.3	0.2	0.7	0.1	20.1***	2.2	0.1	8.8**	0.0
Perennial grasses c.a.	0.1	0.3	0.4	33.1**	13.5**	3.1	84.1***	11.9**	0.1	27.4***	9.8**	0.1
Species richness	5.1*	4.9*	0.5	0.0	18.9***	0.1	32.9***	76.2***	0.0	7.3*	24.9***	2.1

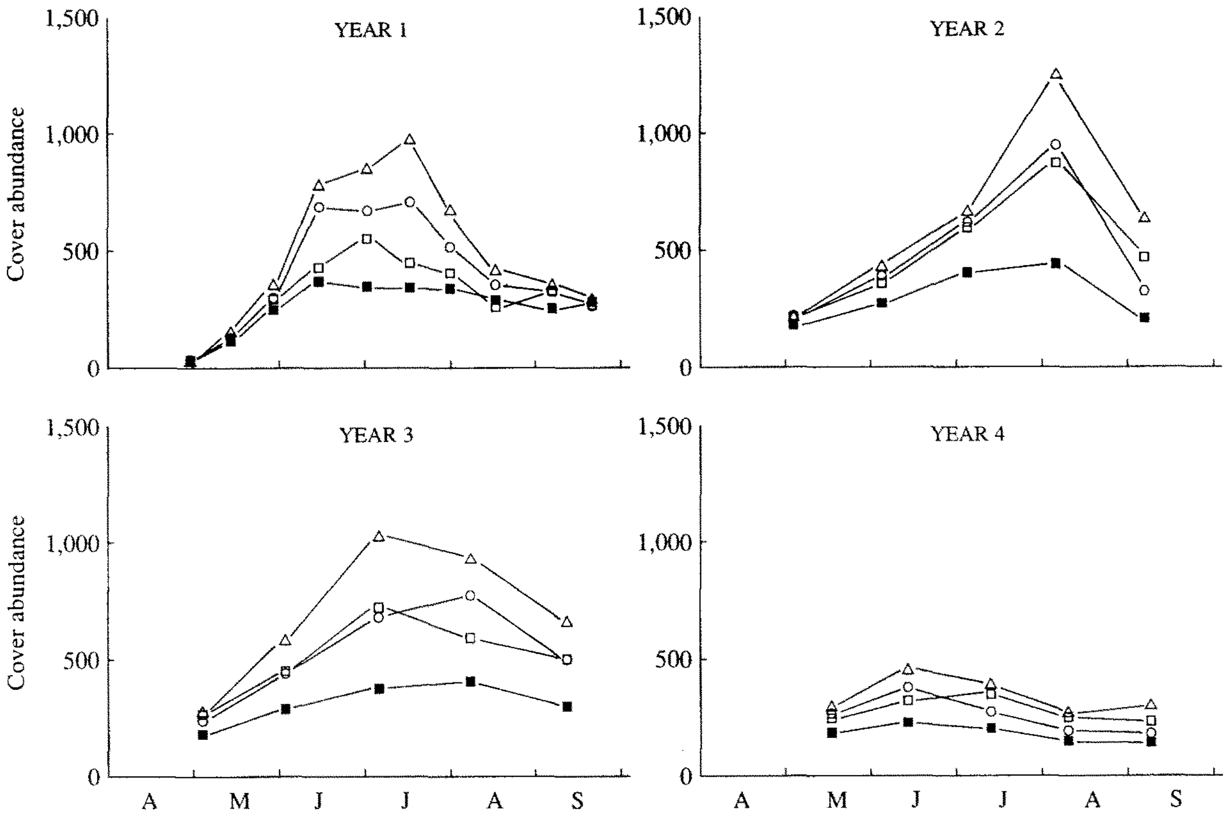

Fig. 2. Trends in cover abundance of a plant community during the first four years of succession. (--■--) control, (--○--) application of foliar insecticide, (-□-) application of soil insecticide, (--△--) application of both compounds.

Perennial forbs became established during the second year of the succession and were unaffected throughout by the application of foliar insecticide (Fig. 4; Table 1). However, during the third and fourth years, there was a significant increase in the cover abundance of this grouping as a result of the application of soil insecticide (Fig. 4c,d; Table 1). Perennial grasses were also very rare during the first year, but became established during the second year. In this and in years three and four, their cover was significantly enhanced by both insecticides (Table 1), although the effect of the foliar insecticide was more pronounced. Indeed, during the second to fourth years of the succession, changes in the overall community cover abundance in foliar insecticide-treated plots were largely a reflection of the dramatic increase in the perennial grass growth (Fig. 2; Fig. 5). In soil insecticide-treated plots, perennial grasses did not contribute as much as to the total vegetation cover (Fig. 2; Fig. 5).

In addition to changes in the cover abundance of the community, there were also differences in the species richness between treatments (Fig. 6). Application of foliar insecticide significantly increased species number in the first year, when more annual forbs became established (Fig. 6a). In the second year of treatment, there was no effect of foliar insecticide application as species richness began to fall, as it did in the control plots (Fig. 6b). Moreover, this decline was continued in years three and four, resulting in a significant reduction in species richness (Fig. 6c,d). In contrast, application of the soil insecticide increased the number of plant species during all four years (Fig. 6). These differences were most marked during years two and three (Fig. 6b,c). During the third year, there was a seasonal average of 17.8

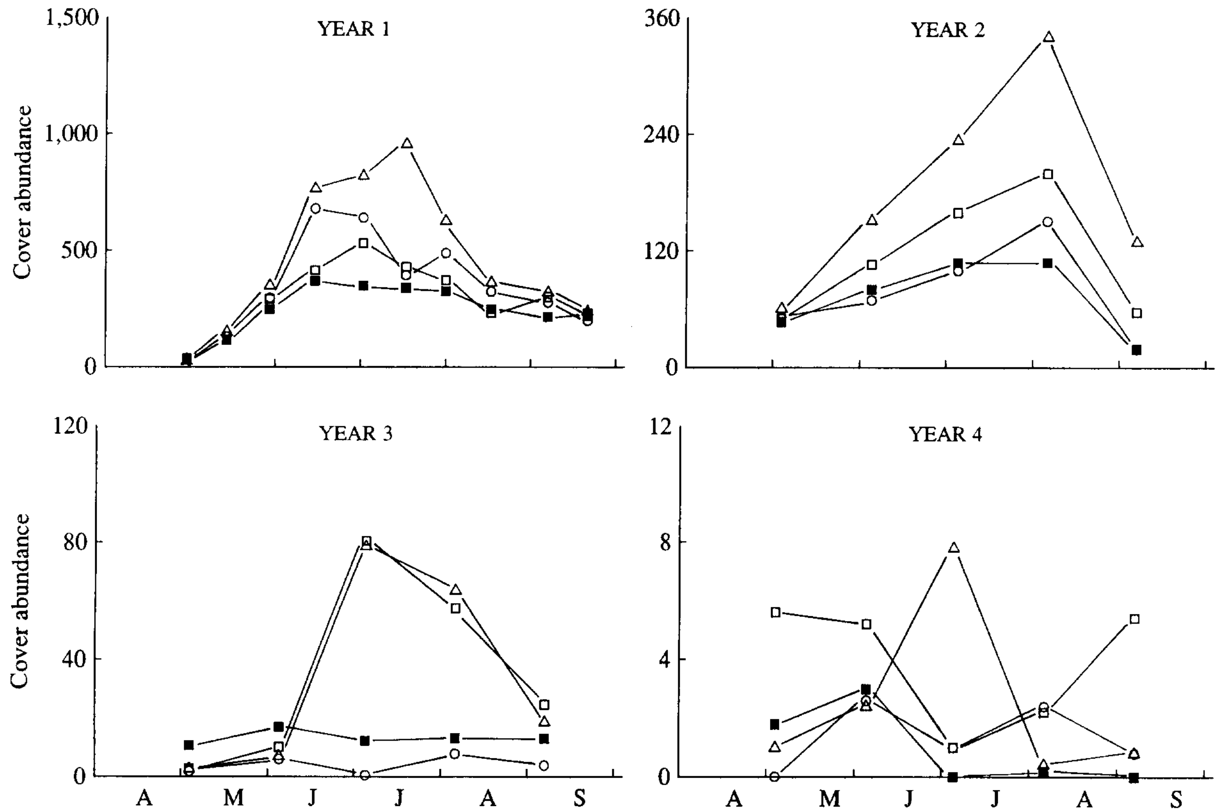

Fig. 3. Trends in the cover abundance of annual forbs during the first four years of succession. Conventions as in Figure 2.

species per plot in the soil insecticide treatment, compared to 9.3 in the foliar treatment (Fig. 6c). Indeed, a total of 17 plant species were only recorded in plots which received soil insecticide during the four years, while no species were only recorded in foliar-treated plots. Of these 17 species, 13 were perennial forbs. It is likely that this is one reason underlying the enhanced performance of this life-history grouping during the third and fourth years of the succession (Fig. 4c,d). The overall decline in species richness in the fourth year was a result of the drought.

Figure 7 summarises the relative effects that the two types of herbivores may have on an early successional plant community, shown in terms of the differences in cover abundance of the major life-history groupings. The greatest effect of foliar-feeding insects was to restrict perennial grass growth, since there were dramatic increases in this life-history grouping when these insects were reduced. Meanwhile, the main effect of root-feeding insects was to restrict the growth of annual and perennial forbs; these groups persisting for longer in the successional sequence when the insects were reduced by the application of soil insecticide. The consequences of herbivore reduction generally increase as the age of the successional community increases.

Discussion

The density of subterranean insect herbivores increases with the successional age of the habitat. This is likely to be a reflection of their long life cycles and relatively poor adult dispersal, which makes them relatively slow colonisers of newly-created, ruderal plant communities (Brown & Gange 1990). Their populations tend to build up slowly and reach their highest densities in mature

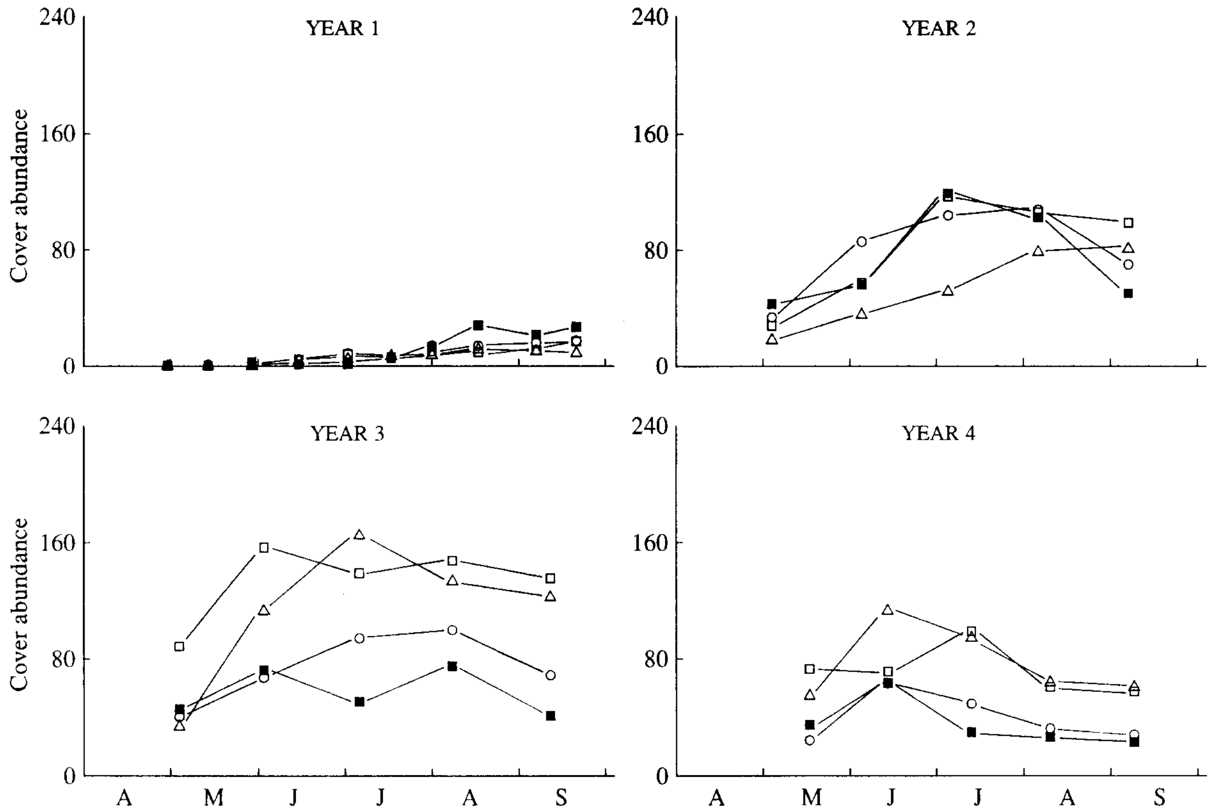

Fig. 4. Trends in the cover abundance of perennial forbs during the first four years of succession. Conventions as in Figure 2.

grassland, characteristic of mid succession (East & Willoughby 1983; Clements *et al.* 1987). In addition, their distributions are often highly aggregated (Gange *et al.* 1991) leading to large standard errors associated with the sample means, such as those recorded here. The proportional composition of the various herbivorous taxa in early succession shows that, in direct contrast to the foliar herbivores, it is the root chewers which are more important than the sap feeders (c.f. Hendrix *et al.* 1988). However, the sap feeders, which are mainly aphids, do become more important in mid successional grasslands.

Despite the relatively low densities of root-feeding insects during the ruderal phase of plant community development, there is still a marked effect on the community when their numbers are reduced by the application of chemicals. In the first year of succession, both root- and foliar-feeding insects reduce the colonisation of annual forbs and, as a result, decrease overall cover abundance and species richness. However, in subsequent years, the effects of these insects are quite different and results such as these serve to emphasise the importance of long-term studies in plant community development (Franklin 1989). It has been shown previously that the predominant foliar-feeding insects in early succession are associated with perennial grasses (Edwards-Jones 1988), and thus it is understandable that a reduction in their numbers leads to a dramatic increase in perennial grass growth. By the third year of succession, perennial grasses dominated the plots where foliar insecticide was applied and their growth tended to suppress that of the competitively-inferior forbs, with the result that species richness was reduced. In addition, the dense grass canopy resulted in fewer available microsites which are necessary for germination and establishment of forb species (Grubb 1977). Therefore,

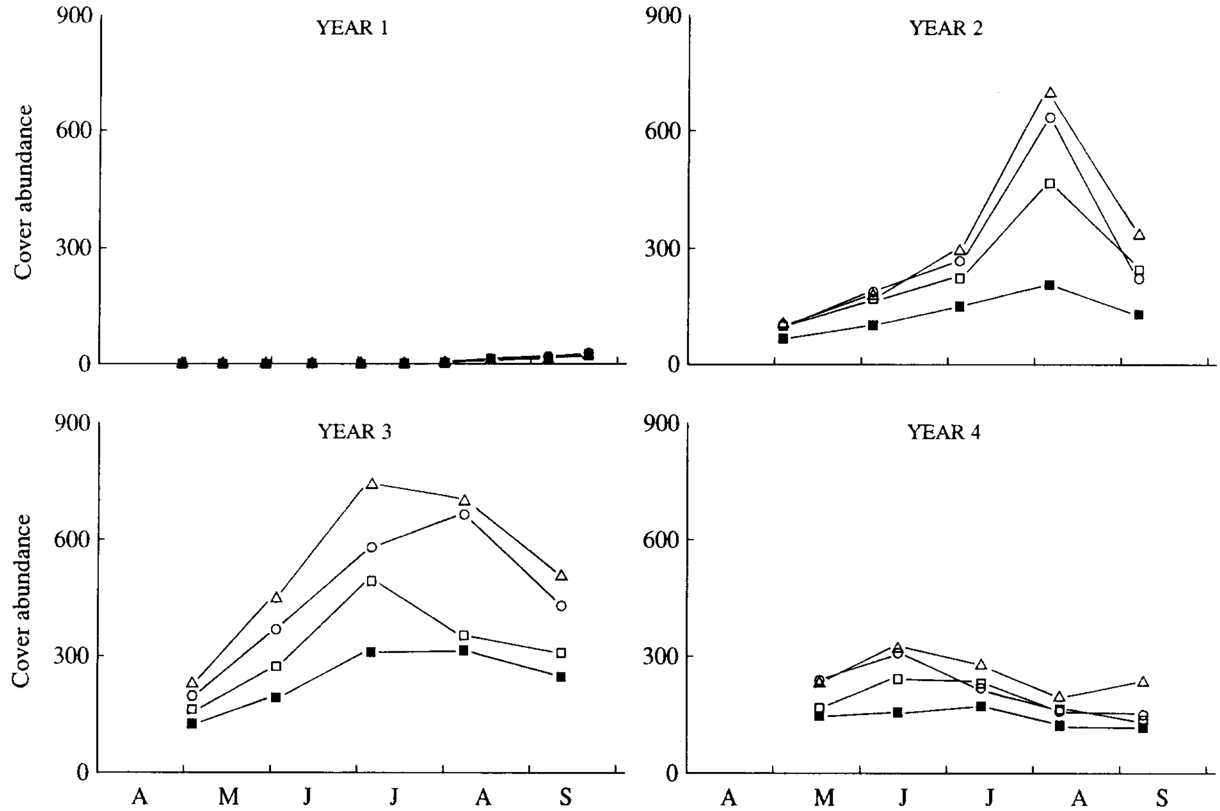

Fig. 5. Trends in the cover abundance of perennial grasses during the first four years of succession. Conventions as in Figure 2.

in early successional grasslands, foliar-feeding insects restrict perennial grass growth, enabling forbs to persist for longer. In this way, they cause a slowing of the rate of succession to a grass-dominated sward and thus resemble the well-documented effects of vertebrate herbivores (Bakker *et al.* 1987; Gibson *et al.* 1987).

However, the effects of root-feeding insects during early succession appear to be different to their foliar-feeding counterparts. Although reducing their abundance does lead to some increase in perennial grass growth, the main effect is to improve the performance of both annual and perennial forbs. Since there were no significant differences in microsite availability (measured as the number of sample pins which touched bare ground) between soil insecticide and control plots, the effects on species richness are likely to be a direct result of the activities of below-ground herbivores. There are few examples of experiments designed to test the effects of above- and below-ground insects on plant species richness. One recent example failed to demonstrate any clear effects (Gibson *et al.* 1990); however these results are hardly surprising since the pesticide treatment employed had few measurable effects on the insect fauna (Evans 1990). In our experiment, 17 plant species only became established where soil insecticide was applied. Such an effect is most likely to result from enhanced seedling establishment in areas where root-feeding insects are reduced. In a parallel experiment, the results of which are to be published elsewhere, we found no differences in the initial spring germination of seedlings in soil insecticide-treated and control plots. However, during the autumn and winter of the first year and spring of the second year, significantly more seedlings appeared in plots where the insecticide was applied. Reducing below-ground herbivory therefore appears to enhance the recruitment of many plant species.

To explore the mechanisms of seedling mortal-

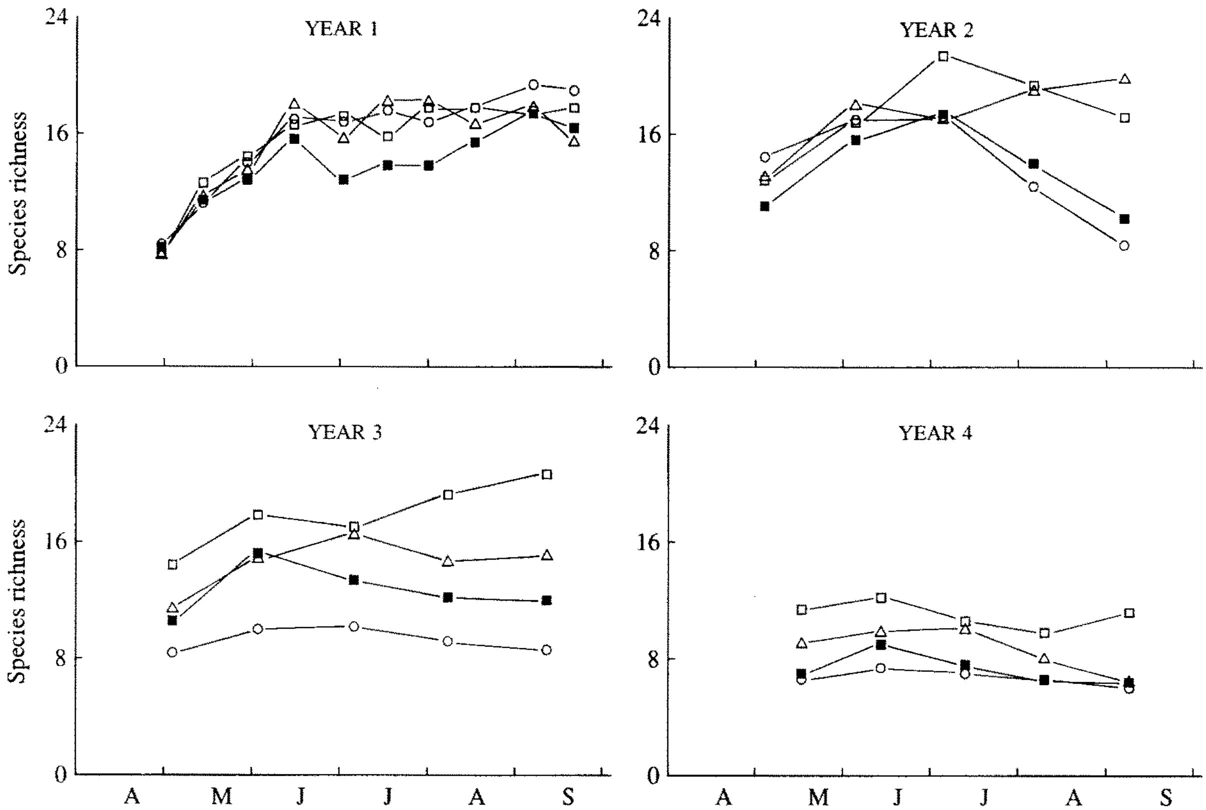

Fig. 6. Trends in plant species richness during the first four years of succession. Conventions as in Figure 2.

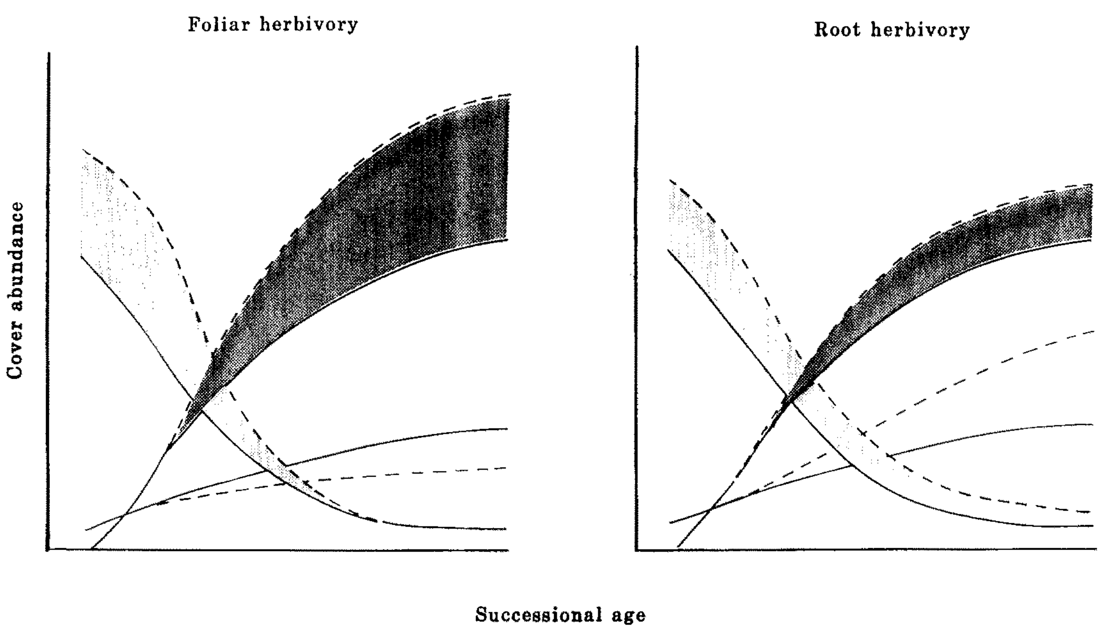

Fig. 7. Model to illustrate the relative effects of above- and below-ground insect herbivory on different plant life-history groupings during early succession. (——) control, (– – –) application of insecticide. (▩) annual forbs, (□) perennial forbs, (■) perennial grasses. Modified from Brown (1990).

ity further, a plot trial using larvae of *Phyllopertha horticola* L. (Garden Chafer) and seeds of *Vicia sativa* L. was set up. It was found that the larvae, at densities equivalent to those in the field, reduced seedling appearance by 60%. Moreover, of the seedlings which died, 85% did so before reaching the soil surface (pre-emergence mortality) (Gange & Brown 1991). These field and laboratory results suggest that subterranean insects are highly important agents of seedling mortality in the field. When their numbers are reduced, greater numbers of plant species are able to recruit and therefore species richness is increased (Brown & Gange 1989b). Thus, in early successional communities, the reduction in recruitment and growth of annual and perennial forbs by subterranean insects will allow enhanced growth of perennial grasses, thereby leading to an increase in the rate of succession. It remains to be seen how subterranean insect herbivores affect later successional plant communities. However, this study revealed that densities of subterranean insects increased with the successional age of the site. Thus, reducing the numbers of these insects in older plant communities may therefore lead to even more dramatic changes in community structure. This is the subject of current experiments at Silwood Park.

These results serve to demonstrate that the composition of successional plant communities, as well as the direction and rate of succession, may be modified by the action of above- and below-ground insect herbivores. The opposing effect of the different modes of insect herbivory enables a balance between the grasses and forbs to be maintained and demonstrates that insect herbivory many indeed be a major determinant in plant community structure and composition.

Acknowledgements

We are grateful to the Natural Environment Research Council for financial support and to Dow-Elanco who generously donated the Dursban.

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